msvtx_nMDT

overview · all regions and selections · rank 8 / 169

rank 8 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.672	0.759	0.603	0.491	0.613	0.767	0.808	0.846	34,970
endcap	ht40	0.654	0.745	0.574	0.457	0.589	0.753	0.795	0.838	7,390
endcap	bkg split	0.056	below 27,580 / above 7,390							
barrel	all	0.331	0.500	0.224	0.177	0.161	0.439	0.555	0.687	1,089
barrel	ht40	0.378	0.547	0.239	0.213	0.193	0.484	0.602	0.729	220
barrel	bkg split	0.061	below 869 / above 220							
all	all	0.560	0.666	0.429	0.366	0.466	0.652	0.708	0.783	36,059
all	ht40	0.543	0.651	0.405	0.343	0.446	0.638	0.694	0.771	7,610
all	bkg split	0.054	below 28,449 / above 7,610							

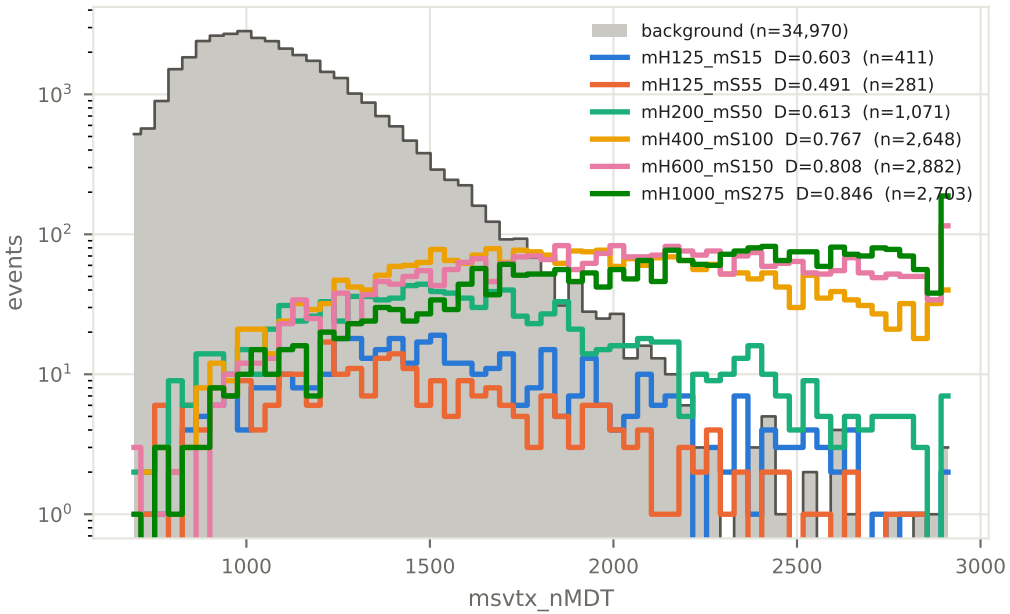
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

msvtx_nMDT

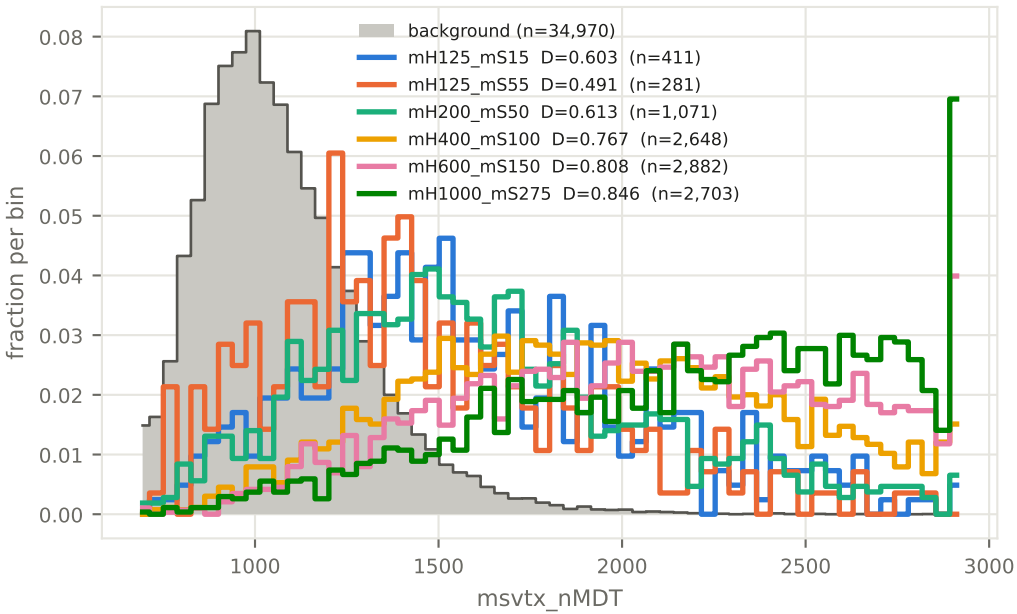
region: ENDCAP · selection: all (every event) · rank 8 / 169

D_equal 0.672 D_pooled 0.759 min 0.491 (mH125_mS55) max 0.846 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

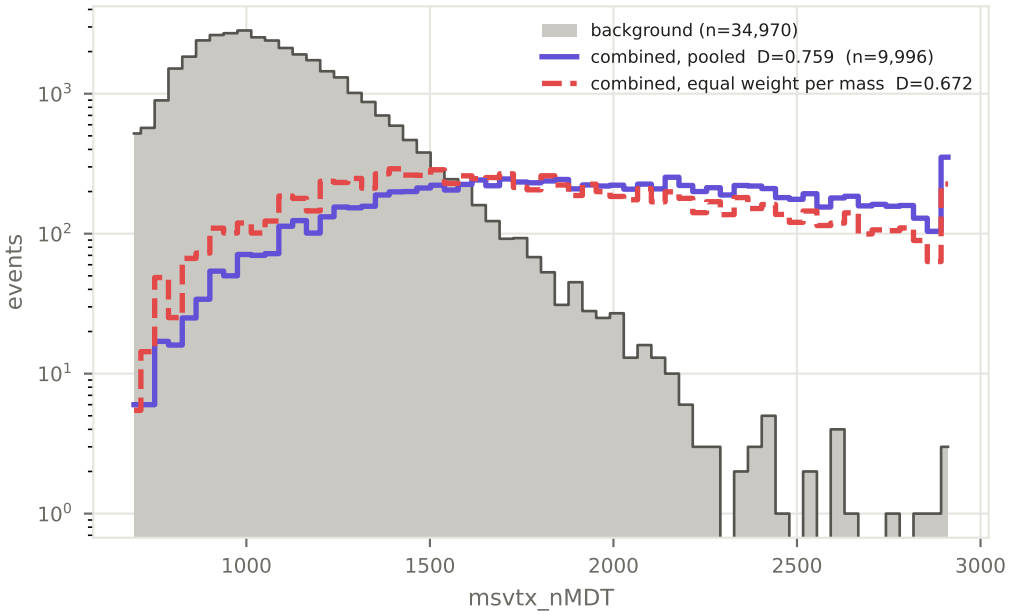
each mass point vs background · raw



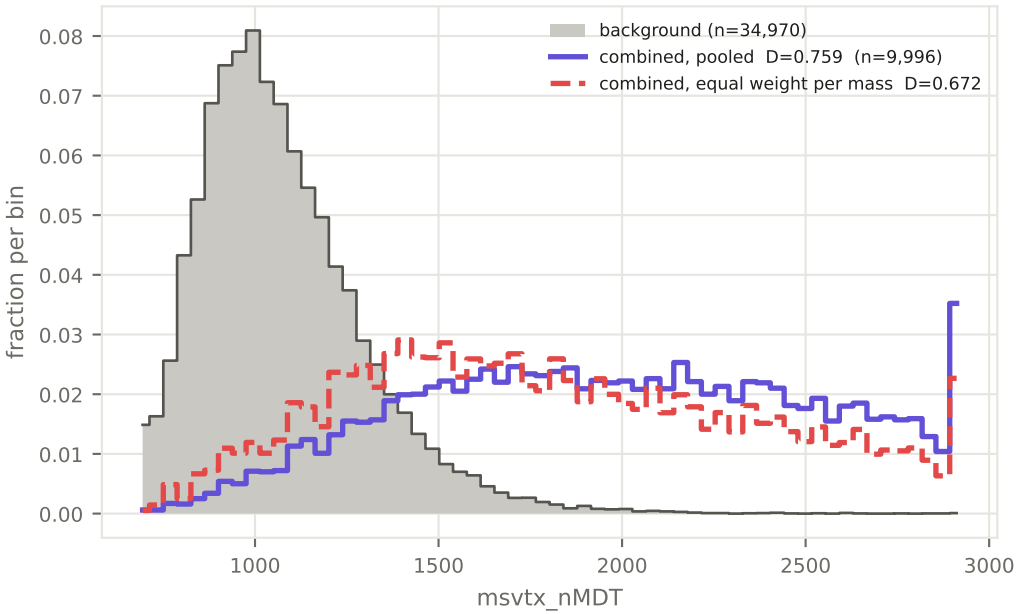
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



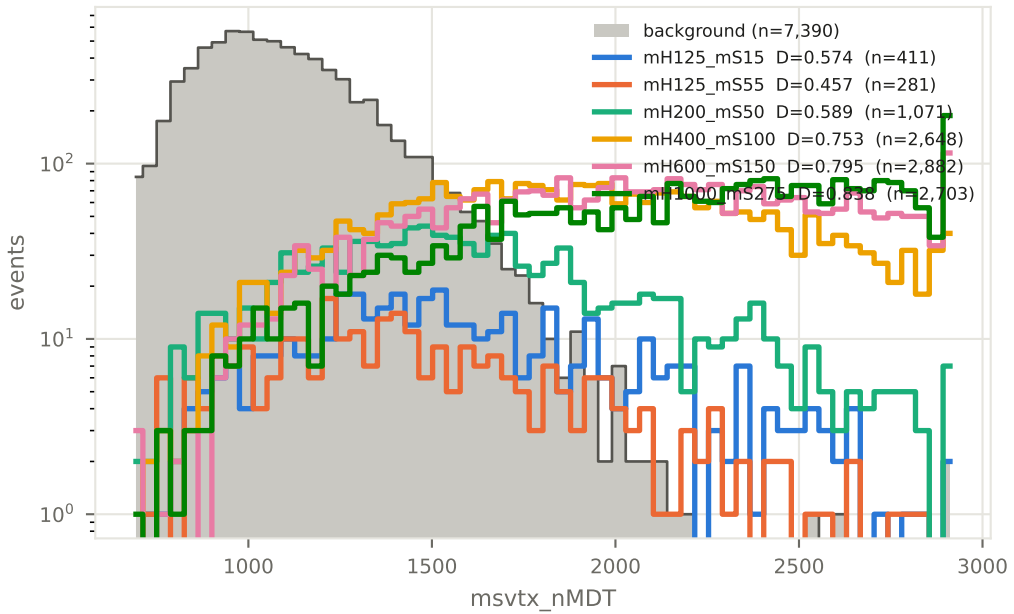
edge bins include under/overflow · NaN and -999 removed

msvtx_nMDT

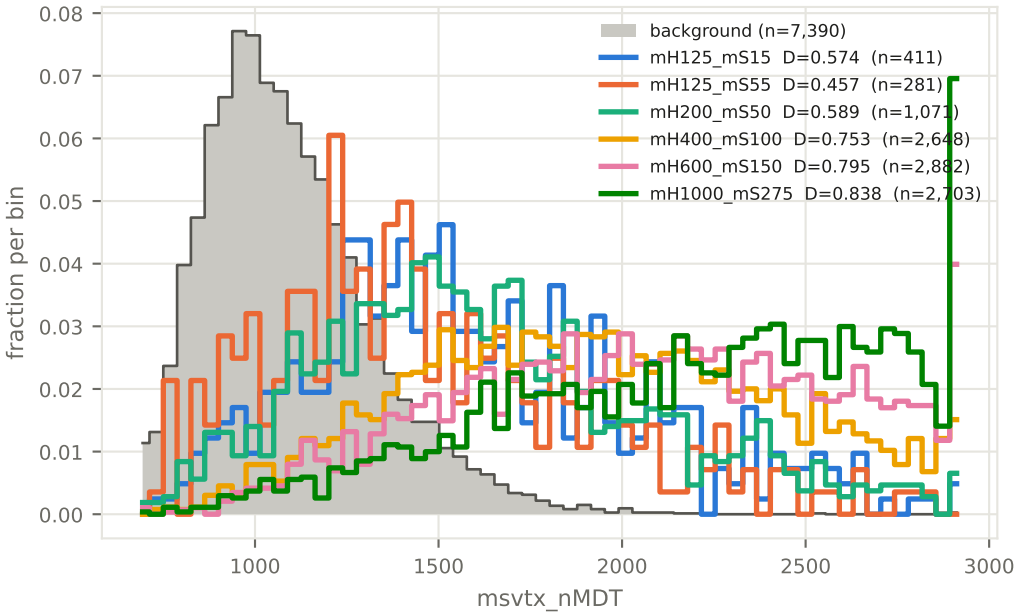
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 8 / 169

D_equal 0.654 D_pooled 0.745 min 0.457 (mH125_mS55) max 0.838 (mH1000_mS275) bkg n = 7,390 sig n = 9,996

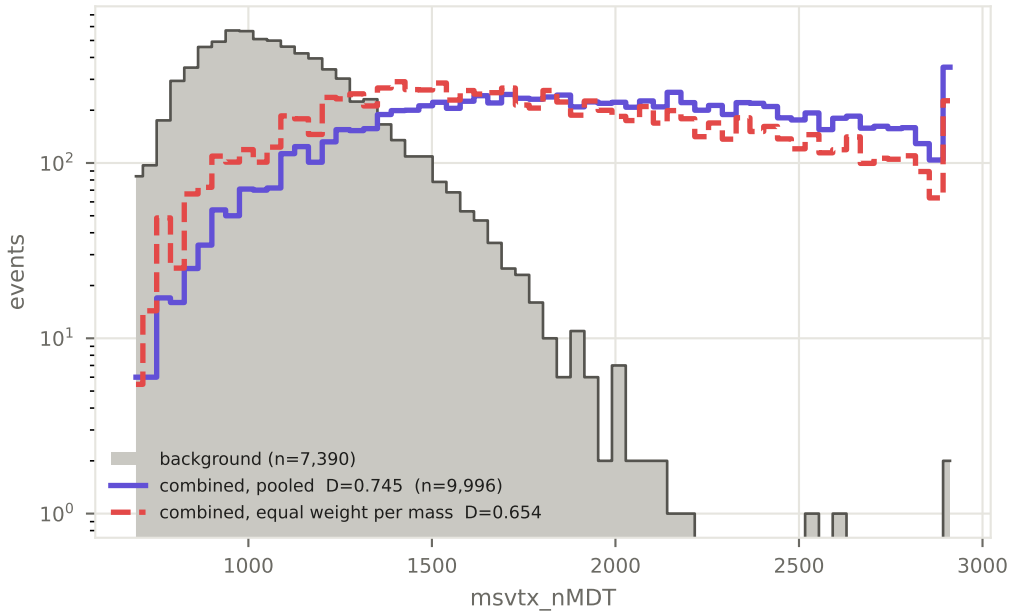
each mass point vs background · raw



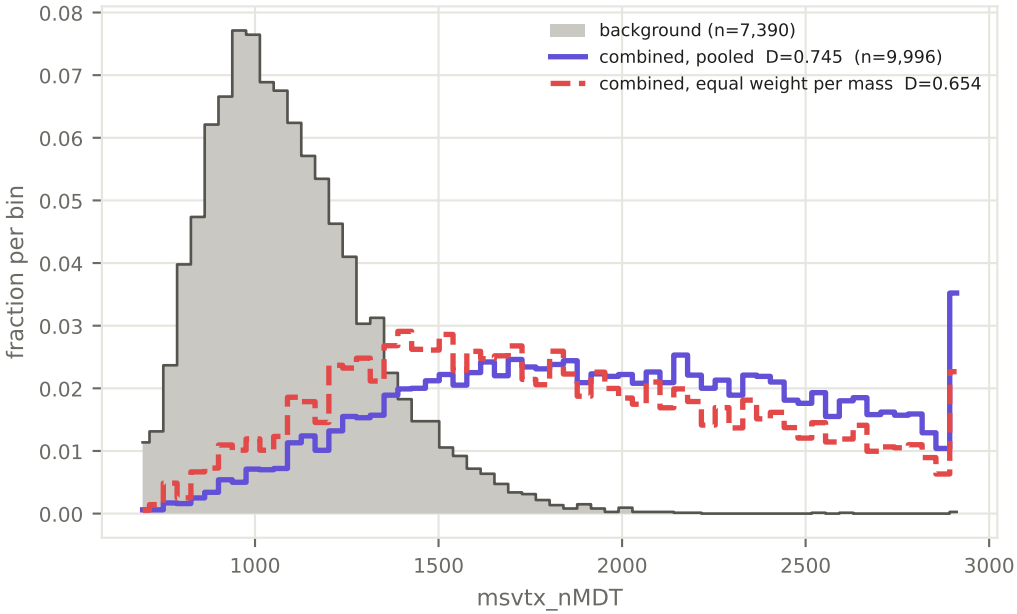
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

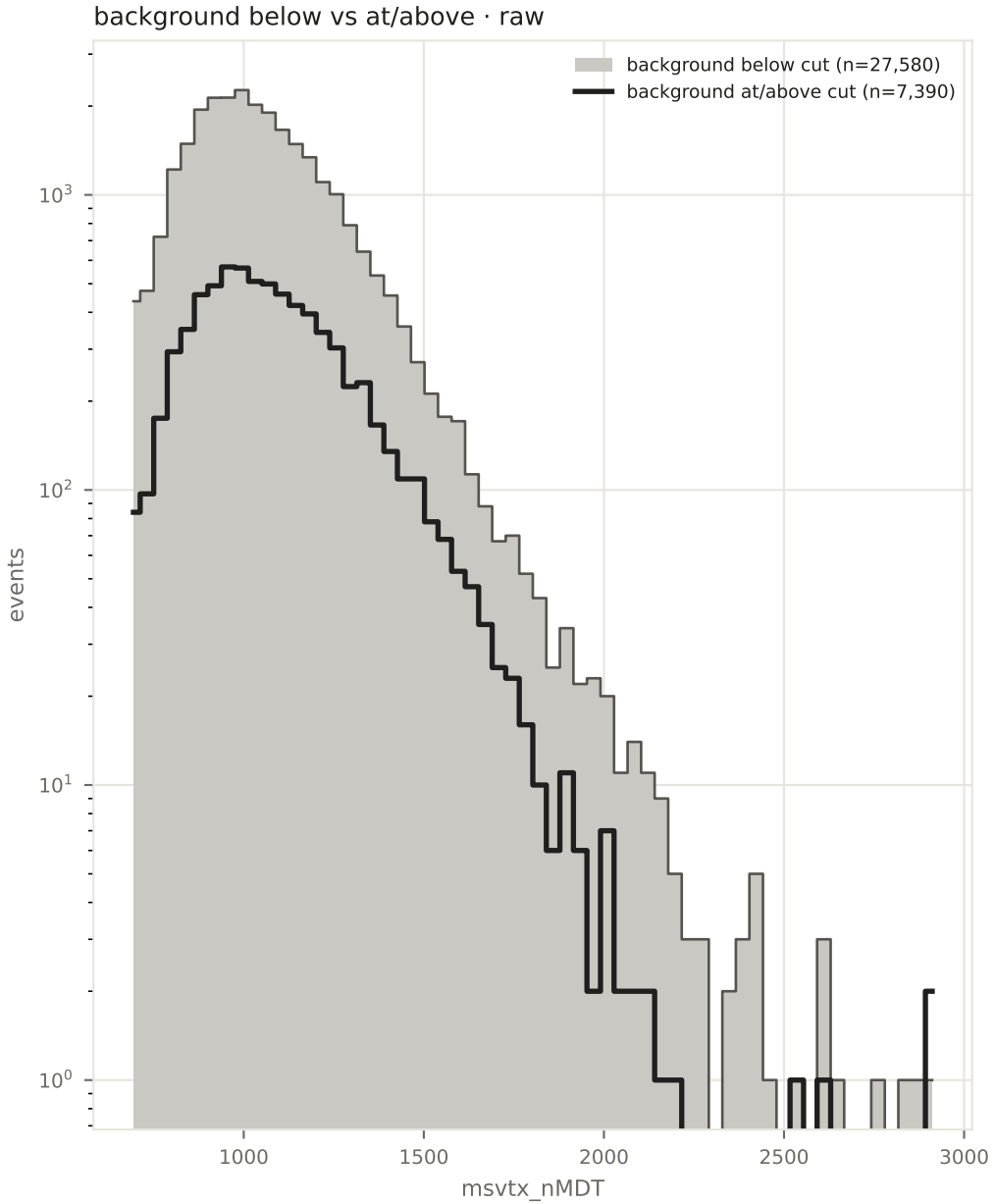


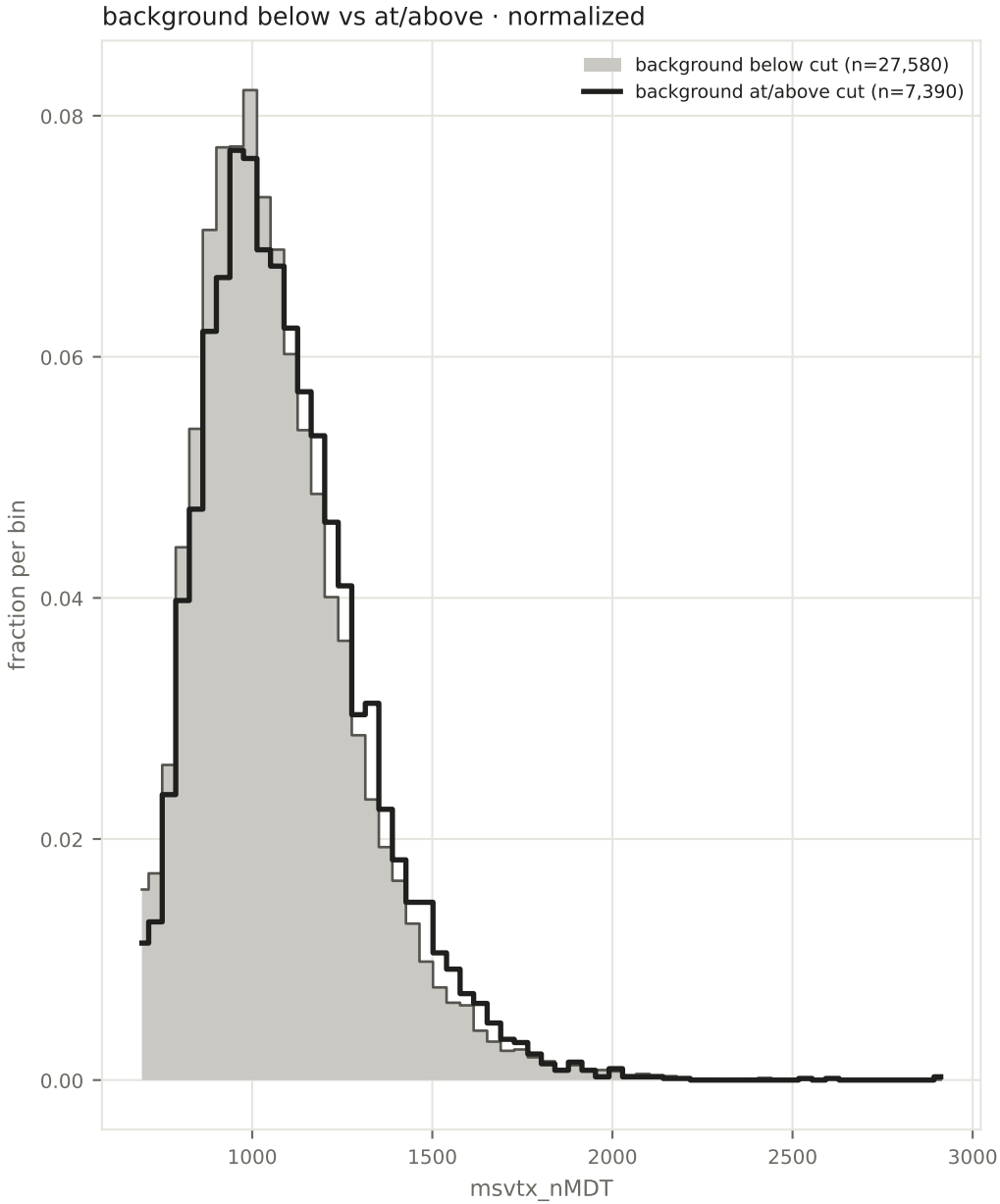
edge bins include under/overflow · NaN and -999 removed

msvtx_nMDT

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 8 / 169

D(below vs above) 0.056 n below = 27,580 n above = 7,390 large D = background shape changes across the cut

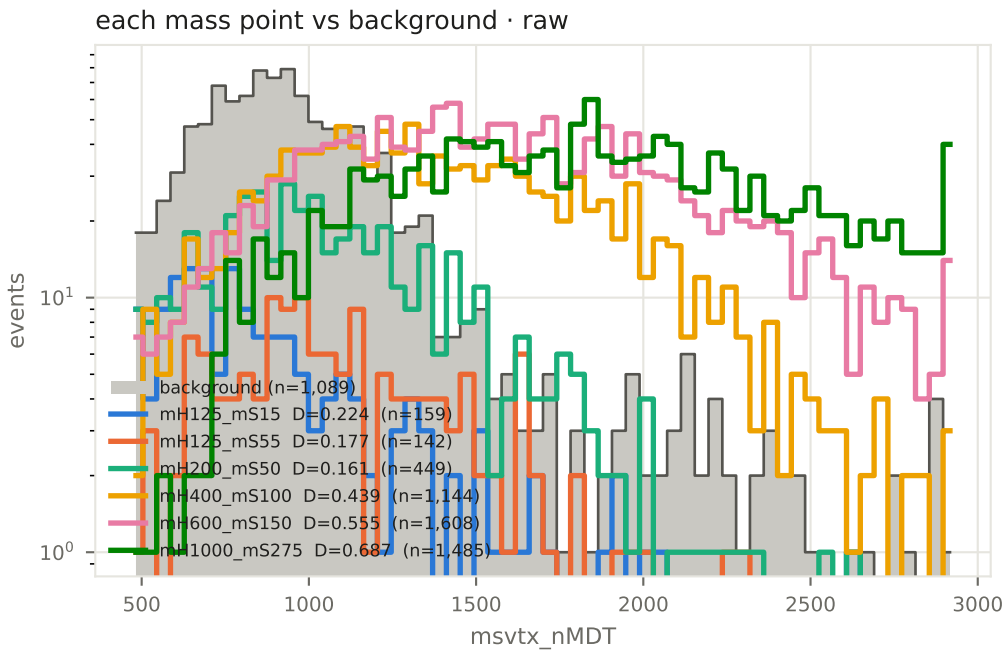


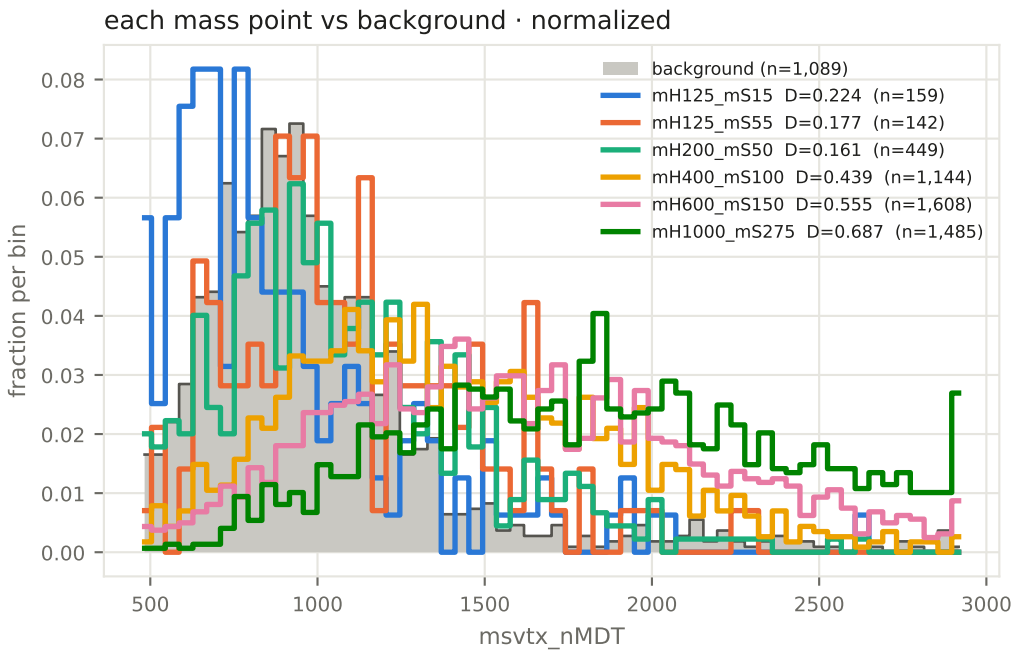


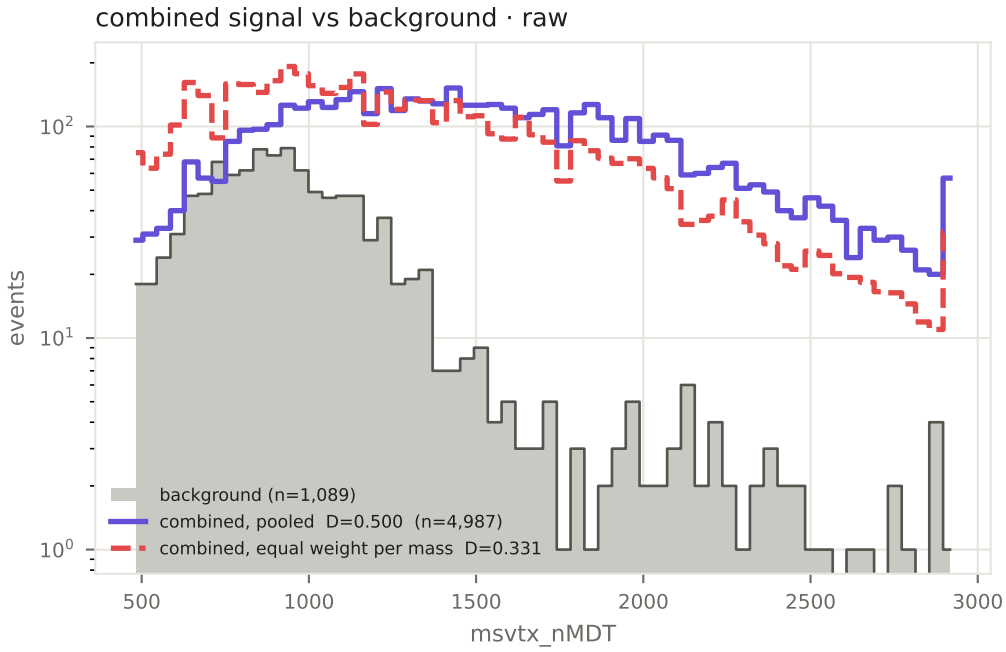
msvtx_nMDT

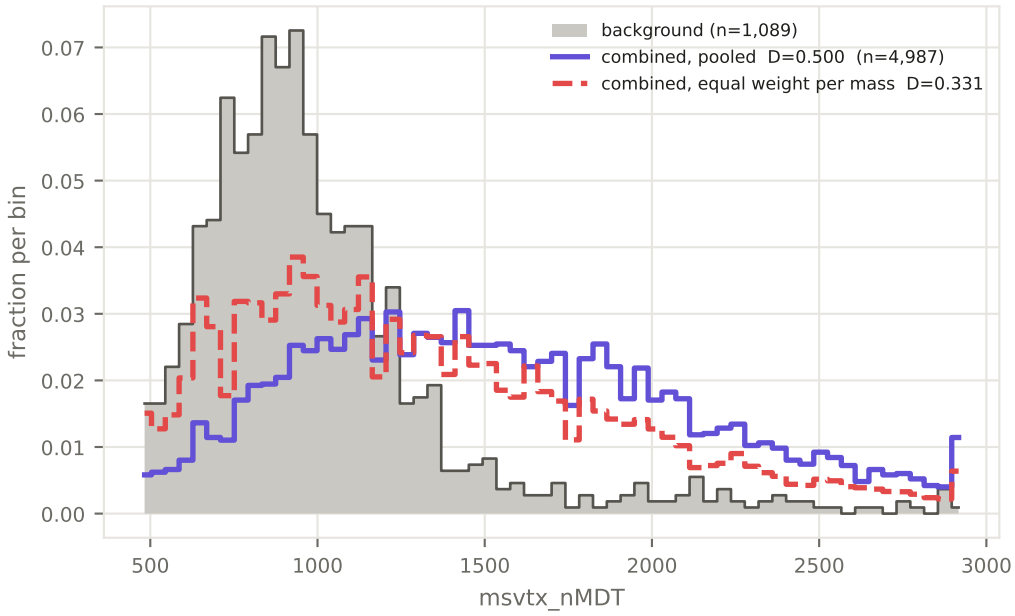
region: BARREL · selection: all (every event) · rank 8 / 169

D_equal 0.331 D_pooled 0.500 min 0.161 (mH200_mS50) max 0.687 (mH1000_mS275) bkg n = 1,089 sig n = 4,987









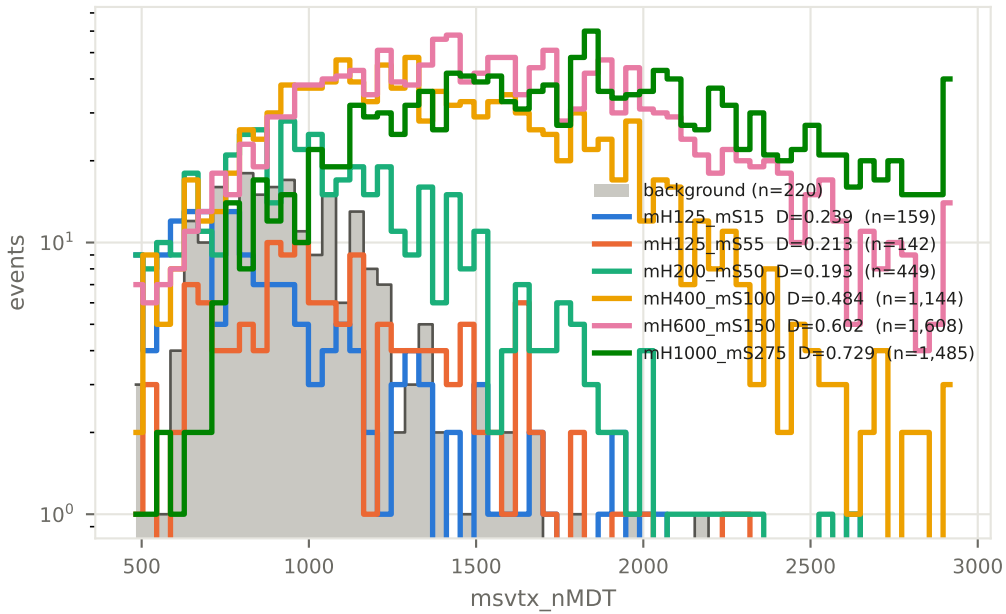
edge bins include under/overflow · NaN and -999 removed

msvtx_nMDT

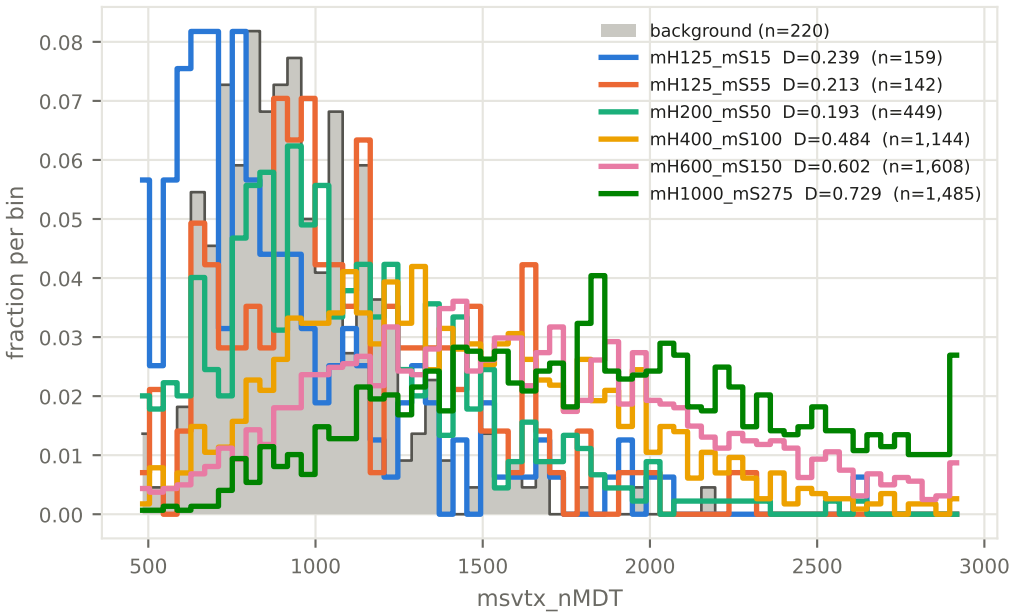
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 8 / 169

D_equal 0.378 D_pooled 0.547 min 0.193 (mH200_mS50) max 0.729 (mH1000_mS275) bkg n = 220 sig n = 4,987

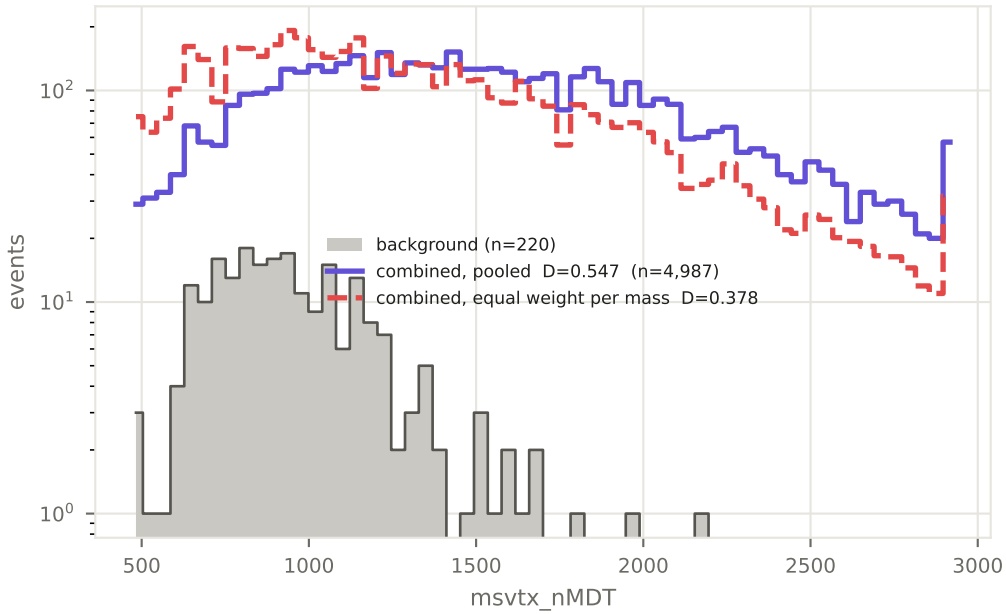
each mass point vs background · raw



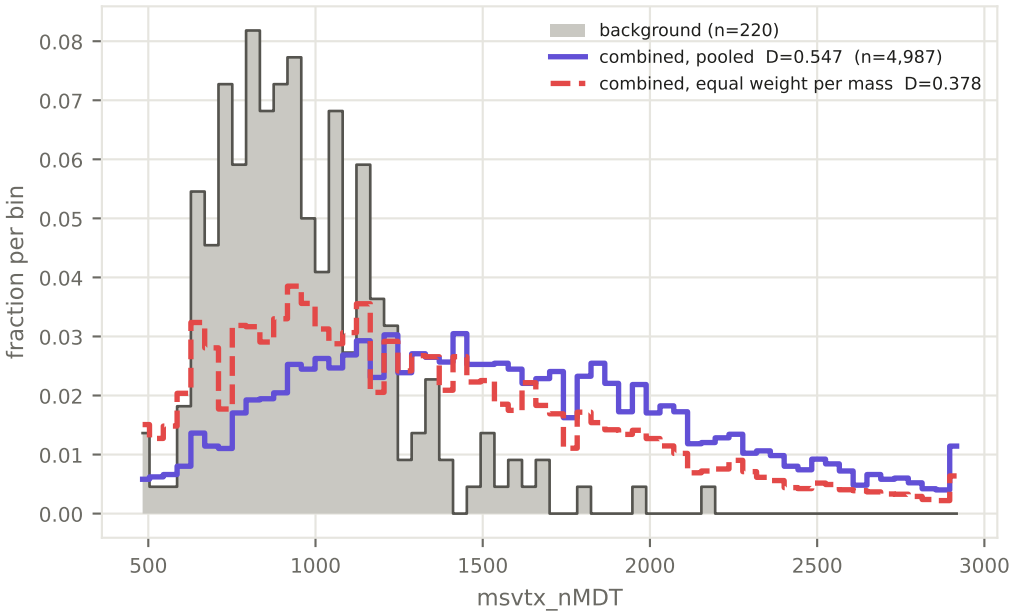
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

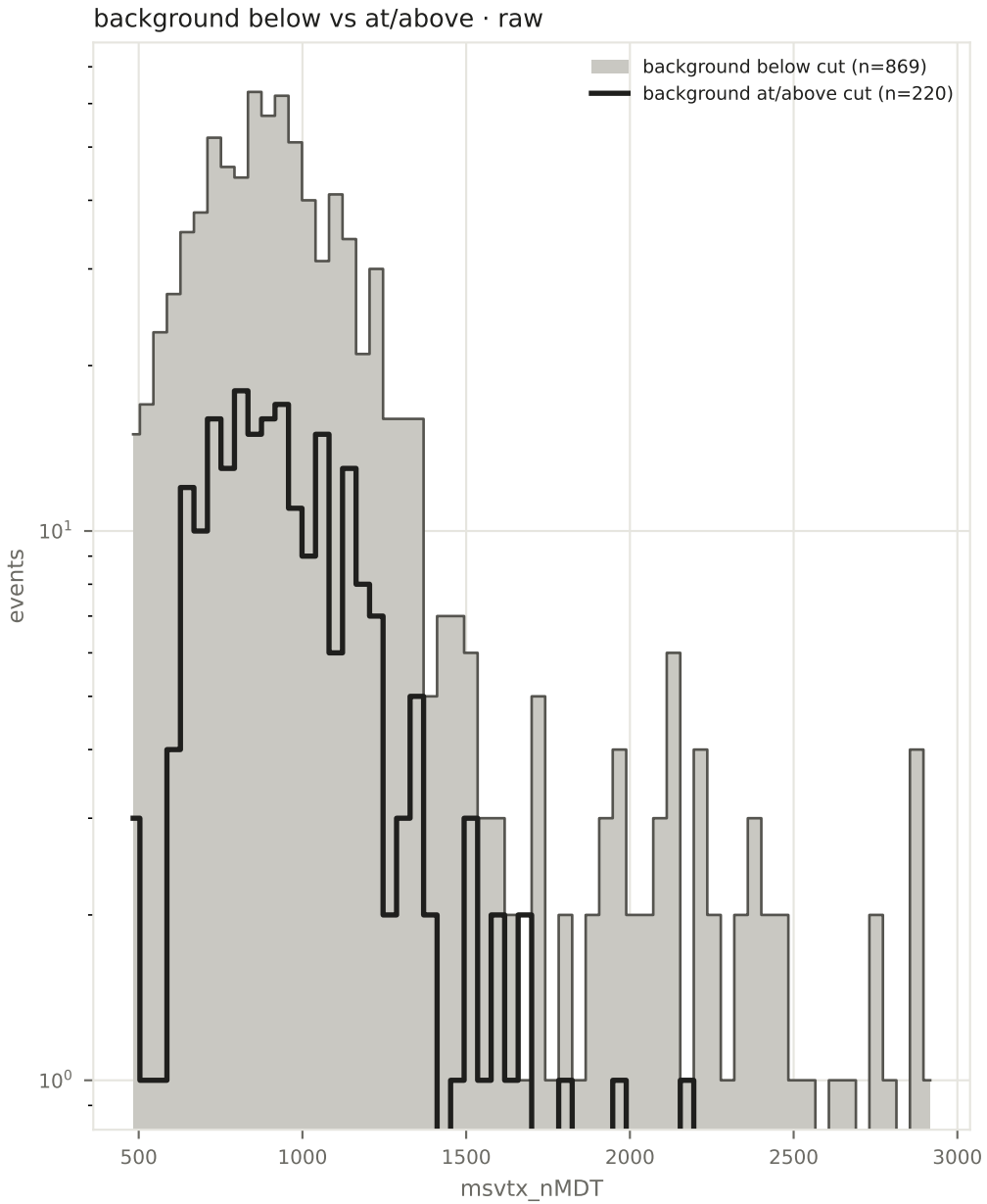


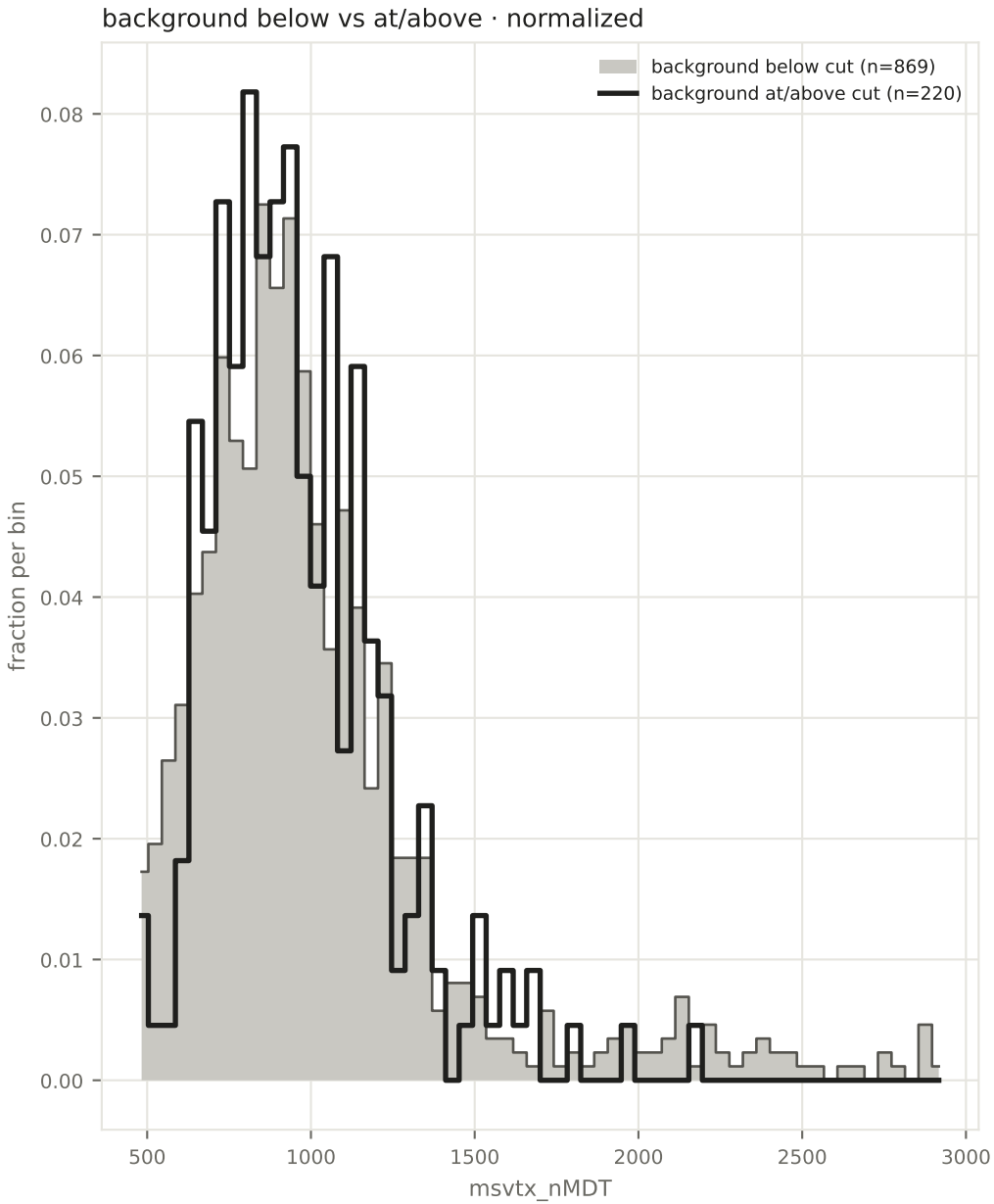
edge bins include under/overflow · NaN and -999 removed

msvtx_nMDT

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 8 / 169

D(below vs above) 0.061 n below = 869 n above = 220 large D = background shape changes across the cut



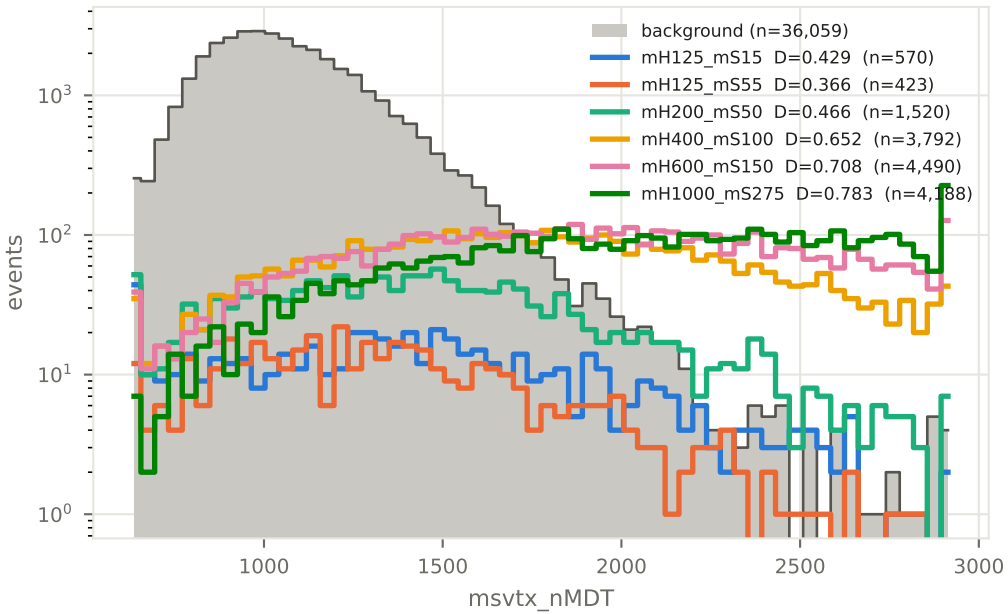


msvtx_nMDT

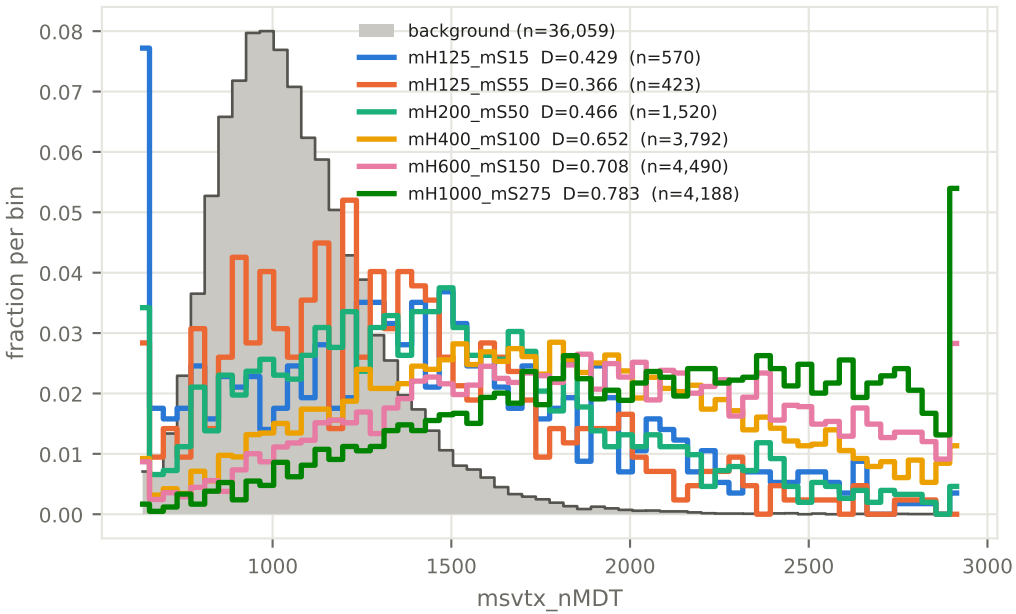
region: ALL · selection: all (every event) · rank 8 / 169

D_equal 0.560 D_pooled 0.666 min 0.366 (mH125_mS55) max 0.783 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

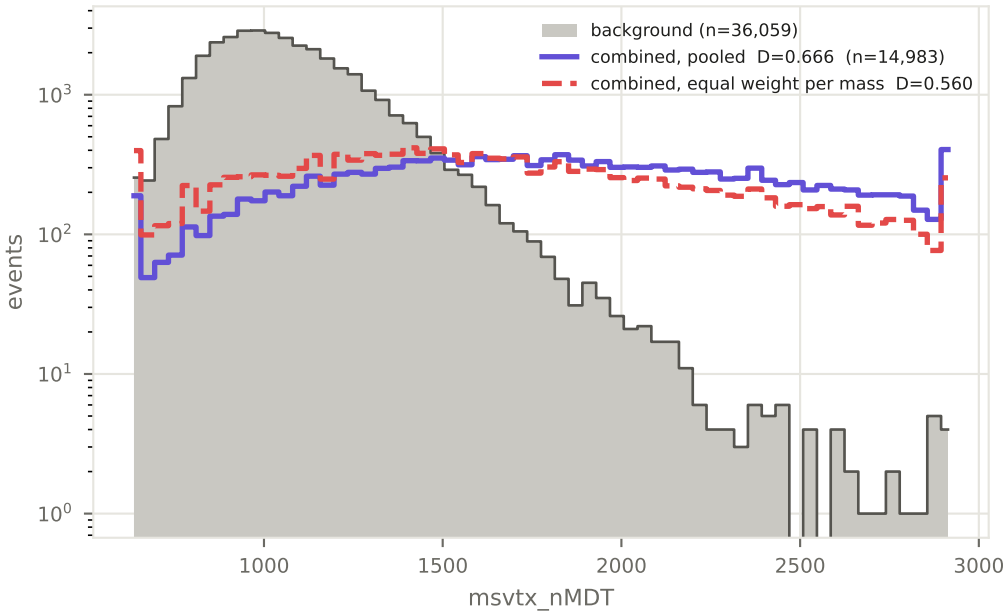
each mass point vs background · raw



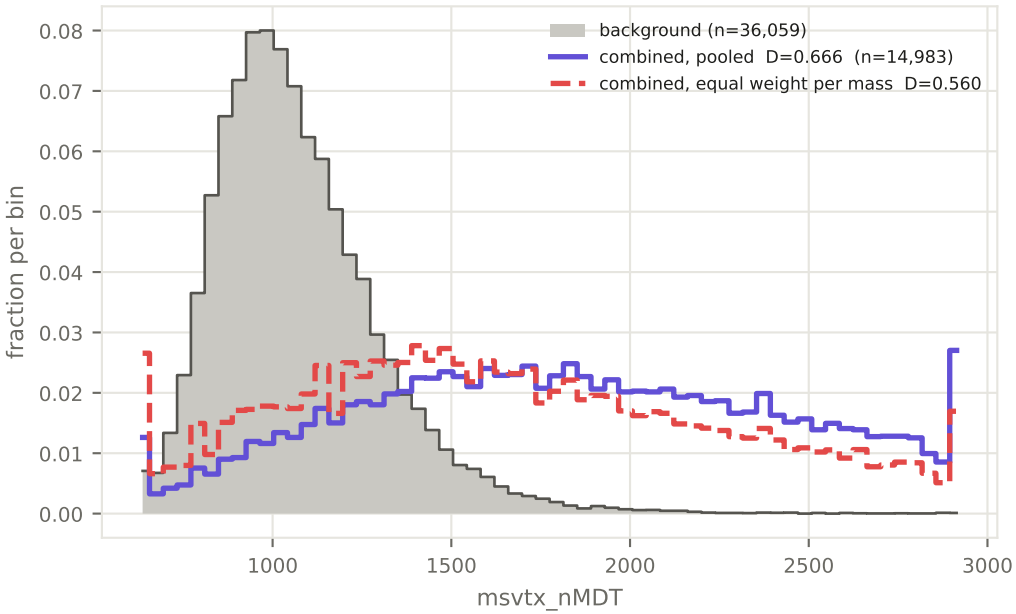
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



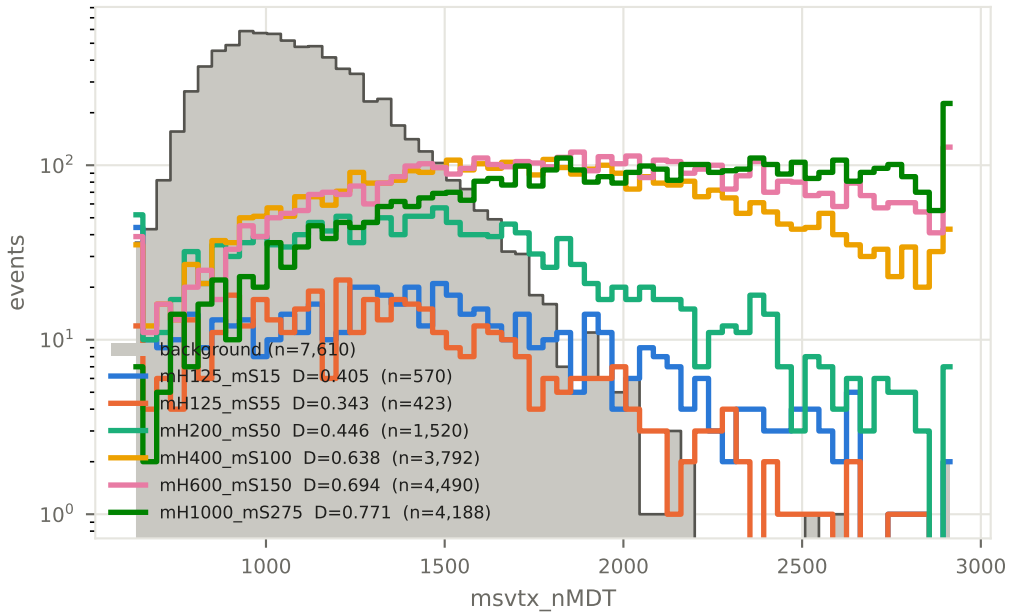
edge bins include under/overflow · NaN and -999 removed

msvtx_nMDT

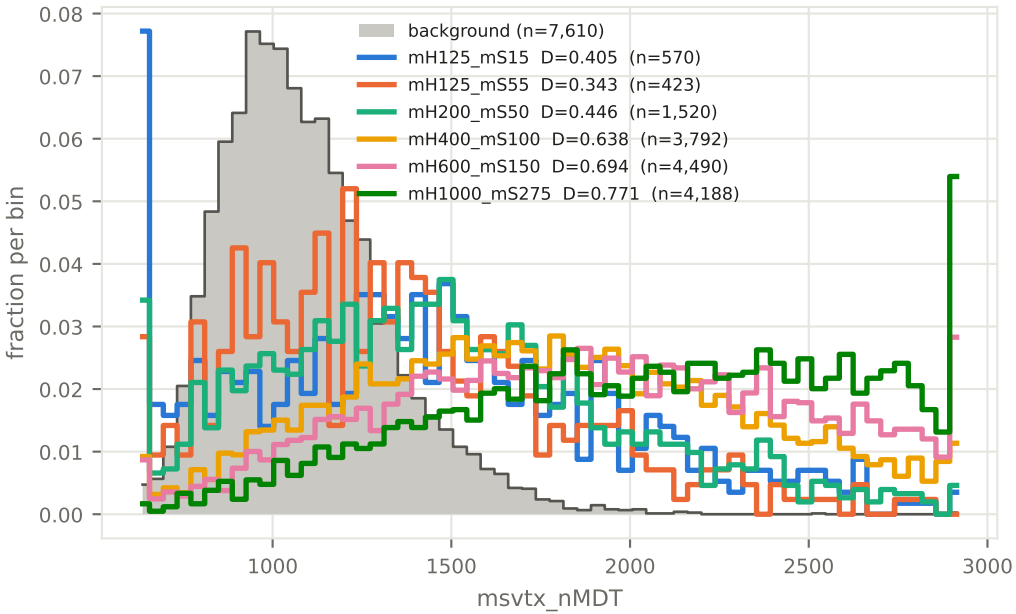
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 8 / 169

D_equal 0.543 D_pooled 0.651 min 0.343 (mH125_mS55) max 0.771 (mH1000_mS275) bkg n = 7,610 sig n = 14,983

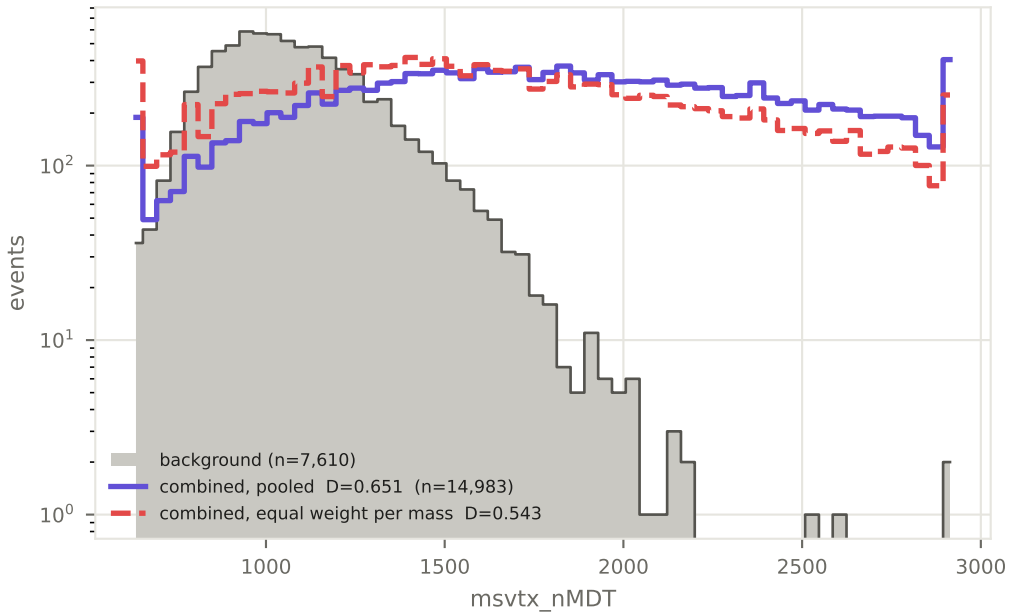
each mass point vs background · raw



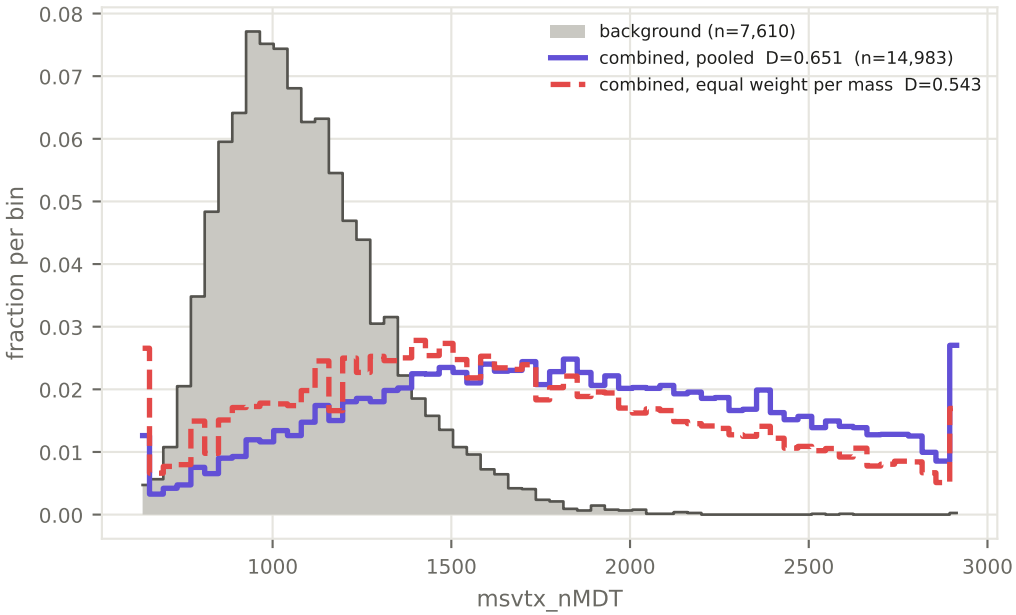
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



edge bins include under/overflow · NaN and -999 removed

msvtx_nMDT

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 8 / 169

D(below vs above) 0.054 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

